2026 IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI) | 979-8-3315-4970-1/26/$31.00 ©2026 IEEE | DOI: 10.1109/IATMSI68868.2026.11465828

# Comparative Power Analysis of Adder Circuits for Low-Power Digital Design

Gitika
Electronics and communication engineering
*Indira Gandhi Delhi Technical University for Women*
New Delhi , India
gitikasingh2002@gmail.com

Prof. Greeshma Arya
Electronics and Communication Engineering
*Indira Gandhi Delhi Technical University for Women*
New Delhi, India
greeshmaarya@igdtuw.ac.in

***Abstract*— As VLSI technology keeps getting smaller over time, adders have become a crucial part of practically every modern integrated circuit. An adder must be fast, with efficiency in terms of power consumption and chip area coming in second. As more compact digital devices with the lowest power consumption continue to emerge and replace one another, the competition to design the most optimal device structure is becoming more intense and quicker. For modern digital systems to improve performance, the ideal adder architecture is essential. This study examines five 4-bit adder architectures from the wide field of adders: Ripple Carry Adder (RCA), Carry Look Ahead Adder (CLA), Carry Select Adder (CSA), Carry Save Adder (CSaA), and Carry Skip Adder (CSkA). The study assesses adder designs' performance characteristics in terms of total on-chip power consumption using Xilinx VIVADO software. The study shows that the Ripple Carry Adder (RCA) consumes the highest power at 5.072 W, primarily due to its sequential carry propagation. The Carry Save Adder (CSA) consumes less power at 4.494 W but is still higher than the more advanced adder designs. The Carry Select Adder (CSA), Carry Lookahead Adder (CLA), and Carry Skip Adder (CKA) demonstrate significantly lower power consumption, with CLA and CKA being almost identical at 2.955 W, reflecting their optimized architectures. Analysis of static and dynamic power indicates that RCA has higher leakage currents, while CLA and CKA minimize dynamic power through reduced switching activity. Overall, for high-performance and low-power applications, CLA or CKA are the preferred choices, whereas RCA, despite its simplicity, is less suitable for energy-sensitive designs.**

***Keywords—Adder, Ripple carry adder, Carry lookahead adder, Carry save adder, Carry select adder, Carry skip adder, Power***

## I. Introduction

Modern integrated circuits (ICs) require high-speed arithmetic units with low power consumption and silicon area due to the ongoing downscaling of VLSI (Very Large Scale Integration) technology. The most basic arithmetic operation is addition, which serves as the foundation for digital systems including processors, digital signal processors (DSPs), and communication circuits

Power dissipation and propagation delay have become significant design issues as technology nodes get smaller. For contemporary VLSI systems, it is crucial to assess and optimize conventional adder architectures due to the need for portable, high-performance, and energy-efficient devices. The need for digital circuits that are faster, smaller, and more energy-efficient has increased as Very Large Scale Integration (VLSI) technology continues to scale down into the deep submicron and nanoscale regimes. Adders, one of these circuits' basic building blocks, are essential to arithmetic and logical operations and have a direct impact on the functionality of microprocessors, digital signal processors, and application-specific integrated circuits (ASICs). Designing adders that strike the best possible balance between speed, power consumption, and silicon area is the main challenge because these factors frequently conflict. Conventional adder architectures, such as Ripple Carry Adder (RCA), are simple and take up little space, but because of sequential carry propagation, they have high power dissipation and extended propagation delays. Conventional adder topologies need to be reevaluated due to the increasing demand for low-power, high-performance systems in embedded devices, mobile computing, and Internet of Things applications. Creating designs that lower dynamic and static power without compromising speed is crucial since power consumption is now a crucial design constraint, particularly in battery-operated systems. A thorough performance comparison in terms of power consumption under equivalent simulation settings is still lacking, despite the fact that a number of fast adder designs, such as Carry Lookahead Adders (CLA), Carry Skip Adders (CSkA), and Carry Select Adders (CSLA), have been presented. In order to identify the best power-efficient and performance-optimized design that uses popular VLSI design tools like Xilinx Vivado, this study's aim is to systematically evaluate and analyze several 4-bit adder architectures.

Since adders are the basic components of arithmetic and logic units (ALUs), efficient adder design has continued to be an important field of study in digital circuit design. in order to optimize power consumption, latency, and area utilization, Sarker et al. [1] compared different 4-bit adder architectures built on conventional CMOS technology. In order to demonstrate how architectural choice affects transistor performance, their study focused on trade-offs between ripple-carry, carry-lookahead, and carry-skip adders. In a similar vein, Anuraj et al. [2] expanded the assessment to bigger bit-width adders (8-, 16-, and 32-bit), offering insights into implementation efficiency and scalability issues in high-bit designs. Together, these studies highlight how crucial architectural optimization is to obtaining balanced design metrics in contemporary CMOS circuits. A thorough analysis of 1-bit full adder cells was published by Hasan et al. [3], who examined performance metrics like power, propagation latency, and transistor count across several logic designs. Their research provides a basis for comprehending trends in scalability in larger adder circuits. In support of these investigations, Patel et al. [4] presented an area-delay-efficient, low-power carry skip adder

979-8-3315-4970-1/26/$31.00 ©2026 IEEE

Authorized licensed use limited to: INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY. Downloaded on October 25,2026 at 13:37:23 UTC from IEEE Xplore. Restrictions apply.

that was tailored for high-performance computing applications and showed better power and delay trade-offs. A thorough comparison of several adder architectures was previously offered by Koyada et al. [5], who identified important factors influencing performance across several logic families. When taken as a whole, these contributions offer a thorough grasp of adder optimization methods, setting the foundation for creating high-speed, energy-efficient arithmetic circuits in contemporary VLSI systems.

## II. Design and Methodology

AMD Xilinx established Xilinx Vivado, a sophisticated integrated design environment (IDE) for digital circuit design and analysis on FPGA and SoC platforms. With improved synthesis, implementation, and simulation capabilities tailored for contemporary programmable logic devices, it replaces the old ISE Design Suite. Vivado enables designers complete control over timing, power, and resource optimization by integrating several design stages—RTL design entry, behavioral simulation, synthesis, implementation, verification, and bitstream generation—into a single environment. Vivado uses the Vivado Synthesis Engine, which reduces power consumption and improves timing closure via a high-level logic optimization approach. The schematic diagram of 4-bit RCA, CSA, CLA, CsaA, CskA are shown in Fig. 1, Fig. 2, Fig. 3, Fig. 4, Fig. 5 respectively. Fuctionality check of each adder is shown in Fig. 6, Fig. 7, Fig. 8, Fig. 9, Fig. 10. Vivado's Power Analysis and Optimization tool is one of its main advantages. By taking into consideration both static (leakage) and dynamic (switching) power components, this feature enables precise estimation of the overall on-chip power usage. For providing comprehensive power breakdowns, the tool takes operational frequency, signal activity, and design specifications into account.

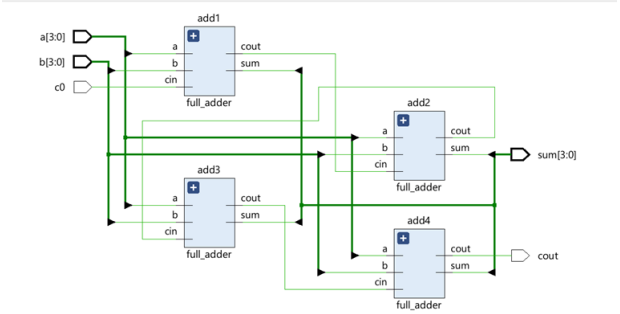



Fig. 1. Ripple carry schematic

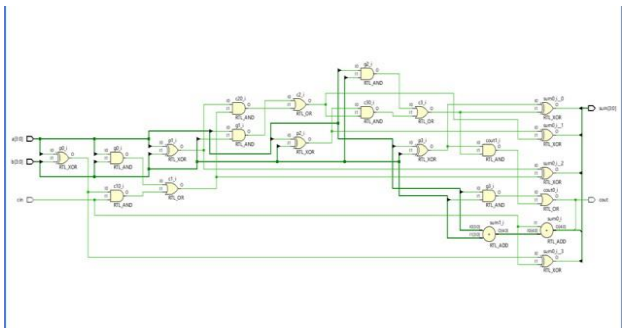


Fig. 2. Carry look ahead adder schematic

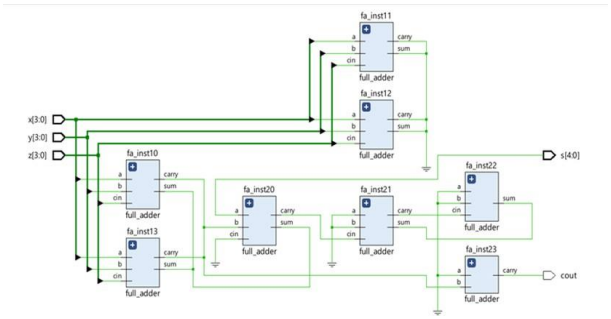



Fig. 3. Carry save adder schematic

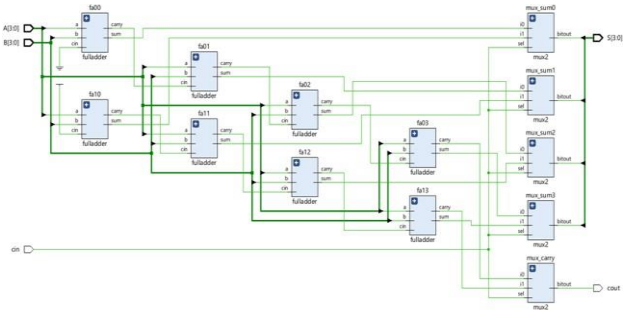



Fig. 4. Carry select adder schematic

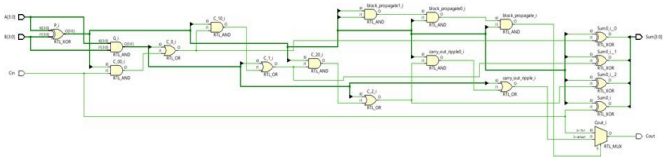


Fig. 5. Carry Skip adder schematic




Fig. 6. Ripple carry waveform

Authorized licensed use limited to: INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY. Downloaded on October 25,2026 at 13:37:23 UTC from IEEE Xplore. Restrictions apply.

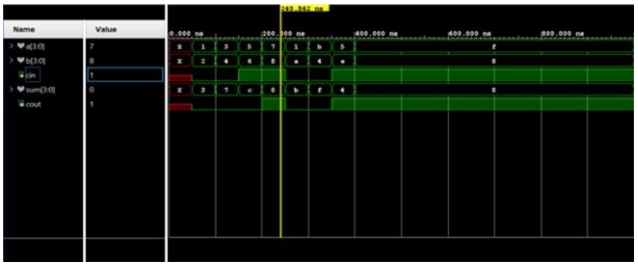


Fig. 7. Carry look ahead adder waveform

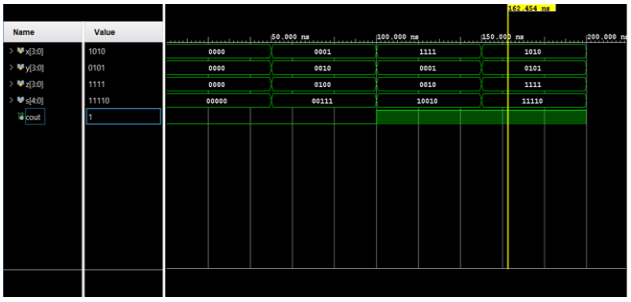


Fig. 8. Carry save adder waveform

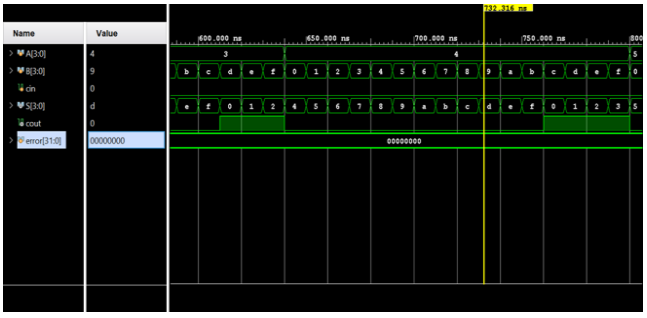


Fig. 9. Carry select adder waveform



Fig. 10. Carry Skip adder waveform

## III. Results and Discussion

To evaluate the power consumption characteristics of the five 4-bit adder architectures—Ripple Carry Adder (RCA), Carry Look Ahead Adder (CLA), Carry Select Adder (CSA), Carry Save Adder (CSaA), and Carry Skip Adder (CSkA)—Xilinx Vivado was used for simulation and analysis. The proposed 4-bit architectures's performance simulation results in terms of power consumed including static and dynamic power are summarized in Table I. Power study provided a comprehensive understanding of each architecture's efficiency by concentrating on both static (leakage) and dynamic power components. The findings show that the Ripple Carry Adder (RCA) has the greatest overall power consumption of about 5.072 W. This is mainly because of its sequential carry propagation method, which produces longer signal routes and more switching activity. The Carry Save Adder (CSaA) still has redundant signal transitions in its partial carry generation process, despite a minor improvement in performance with a total power usage of 4.494 W. On the other hand, due to their parallel carry computation and effective carry bypassing techniques, correspondingly, the Carry Look Ahead Adder (CLA) and Carry Skip Adder (CSkA) exhibit much lower power dissipation, both consuming approximately 2.955 W. Although its dual adder construction results in substantially more power than CLA and CSkA, the Carry Select Adder (CSA) also exhibits lower power consumption compared to RCA and CSaA, benefiting from shorter carry propagation latency. Further investigation of the static and dynamic power components shows that while CLA and CSkA successfully minimize dynamic power through decreased switching activity and optimized logic depth, RCA has greater leakage currents because it has more active nodes. These results imply that the best trade-off between performance and power efficiency is offered by CLA and CSkA architectures. Therefore, CLA or CSkA are the best options for high-speed and low-power applications, while RCA is less effective for energy-constrained conditions despite its structural simplicity and ease of implementation.

TABLE I

TOTAL ON-CHIP POWER CONSUMPTION, STAIC POWER, DYNAMIC POWER

| 4-BIT ADDER | TOTAL ON-CHIP POWER (watt) | STATIC POWER (watt) | DYNAMIC POWER (watt) |
|---|---|---|---|
| RCA | 5.072 | 0.315 | 4.756 |
| CSaA | 4.494 | 0.091 | 4.404 |
| CSA | 2.956 | 0.087 | 2.869 |
| CLA | 2.955 | 0.087 | 2.868 |
| CSKA | 2.955 | 0.087 | 2.868 |

## IV. Conclusion

With Xilinx Vivado simulation tools, this study examined the power efficiency of five 4-bit adder architectures: Ripple Carry Adder (RCA), Carry Look Ahead Adder (CLA), Carry Select Adder (CSA), Carry Save Adder (CSaA), and Carry Skip Adder (CSkA). Understanding how various carry propagation strategies affect overall power consumption in CMOS-based digital systems was the main goal. The comparative analysis showed that optimized architectures like CLA and CSkA achieve much reduced power dissipation through parallel and bypassing techniques, while conventional designs like the RCA demonstrate significantly higher power utilization due to sequential carry propagation. These results highlight the importance of effective carry computation in the development of power-optimized arithmetic circuits. Overall, the study shows that without requiring intricate

Authorized licensed use limited to: INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY. Downloaded on October 25,2026 at 13:37:23 UTC from IEEE Xplore. Restrictions apply.

hardware changes or sophisticated fabrication techniques, architectural advancements can significantly lower power consumption. The knowledge gathered from this work can help designers choose appropriate adder architectures for high-speed, energy-efficient digital applications. To create an improved performance representation, future studies can extend this analysis to higher bit-width adders and include other factors like propagation delay, area utilization, and energy-delay product. The design of low-power, high-performance VLSI systems for next-generation computing technologies might benefit from such a thorough examination.

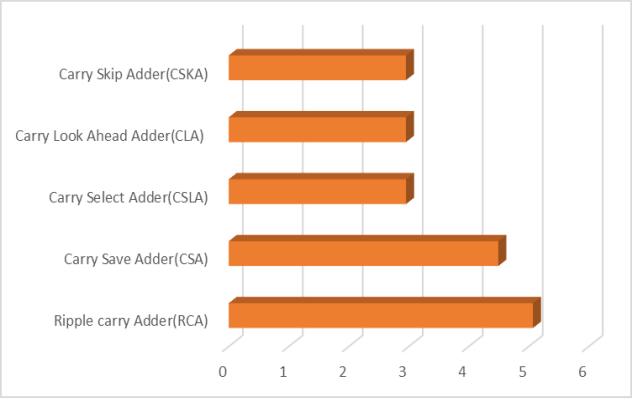



Fig. 11. Graphical representation of total on-chip power consumption

## V. FUTURE SCOPE

Besides the existing 4-bit designs examined in this paper, the ongoing growth of VLSI technology presents interesting opportunities for more adder architecture optimization. In order to assess how power, latency, and area metrics scale with increasing complexity, future study might concentrate on expanding the comparison to larger bit-width adders (8-bit, 16-bit, 32-bit, or 64-bit). Furthermore, using novel low-power design strategies like clock gating, dynamic voltage and frequency scaling (DVFS), and transistor-level optimizations can further lower power consumption while preserving high-speed operation. It would also be beneficial to evaluate these adders' performance utilizing FinFET, GAAFET, or CNTFET technologies in order to comprehend their behavior in next-generation semiconductor nodes, given the increasing importance of developing nanoscale and sub-nanometer technologies. Furthermore, incorporating these architectures into practical applications like embedded systems, digital signal processors (DSPs), and arithmetic logic units (ALUs) may offer greater understanding of useful power-performance trade-offs.

## VI. REFERENCES


[1] Sarker, Anika Tasin, et al. "A Comparative Analysis of 4-Bit Adder Architectures for Enhanced Efficiency in Conventional CMOS Technology." 2024 IEEE International Conference on Power, Electrical, Electronics and Industrial Applications (PEEIACON). IEEE, 2024.

[2] Anuraj, V., Preeti Verma, and Dhandapani Vaithiyanathan. "Implementations and Analysis of Efficient Adders for 8, 16, and 32-Bit Design." 2024 First International Conference for Women in Computing (InCoWoCo). IEEE, 2024.

[3] Hasan, Mehedi, et al. "Comprehensive study of 1-bit full adder cells: review, performance comparison and scalability analysis." SN Applied Sciences 3.6 (2021): 644.

[4] Patel, Sujit, et al. "Area-delay efficient and low-power carry skip adder for high performance computing systems." 2019 IEEE International Symposium on Smart Electronic Systems (iSES)(Formerly iNiS). IEEE, 2019.

[5] Koyada, Bhavani, et al. "A comparative study on adders." 2017 international conference on wireless communications, signal processing and networking (WiSPNET). IEEE, 2017.

[6] Jasbir Kaur and Lalit Sood, “Comparison between various types of adder topologies,” in IJCST, vol. 6, no. 1, Jan.–Mar. 2015.

[7] Maroju SaiKumar and Dr. P. Samundiswary, “Design and performance analysis of various adders using verilog,” in International Journal of Computer Science and Mobile Computing, IJCSMC, vol. 2, no. 9, pp. 128–138, Sep. 2013.


Authorized licensed use limited to: INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY. Downloaded on October 25,2026 at 13:37:23 UTC from IEEE Xplore. Restrictions apply.